

∴ It's easy to see that A can be row reduced to I by row replacements and scaling operations. By Theorem 7, A is invertible. By Defn. of inverse, A^{-1} exists and from Theorem 7, the sequence of row operations that reduces A to I also changes I to A^{-1} .

• At first, scaling is performed on the rows of I . Row i of $I = \frac{1}{a_{ii}} \times \text{old}$ Row i of I . ∴ After performing this operation for each row of I , the main diagonals of I take up the corresponding main diagonal entries of A . The rest of the row entries remain 0.

• For each row replacement operation, the following happens:

$$\cancel{a_{ji}} \quad i_i = i_i - a_{ji} \cdot \frac{1}{a_{ii}} \Rightarrow \cancel{0} = 0 - \frac{a_{ji}}{a_{ii}} \Rightarrow \boxed{i_i = -\frac{a_{ji}}{a_{ii}}}$$

This happens for all $j > i$ and $k < i$.

∴ A^{-1} must be lower triangular since the entries above the main diagonals remain 0.

• From Prob 27(c) and 28 in Exercise 2.2, we know that the sequence of row replacement and scaling operations has equivalent elementary matrices $E_1, E_2, \dots, E_p$ where $(E_p E_{p-1} \dots E_2 E_1) A = I$ and $(E_p E_{p-1} \dots E_2 E_1) I = A^{-1}$. $\square$